

# Andrew Andrusenko

<https://andrewandrusenko.github.io>

<https://github.com/AndrewAndrusenko>

[andrusenko3@gmail.com](mailto:andrusenko3@gmail.com) ☎ 7-(916)-104-5074

FullStack (front/back) Developer | Angular | Node.js | PostgreSQL | Data Visualization | Other..

## SUMMARY - 5 (15) YEARS EXPERIENCE

- I'm a fully committed full stack developer. **I really love what I do.** Creating apps and pushing my limits drives me. **English - C1**
- Main stack is Angular (+RxJs deep), Node.js, PostgreSQL.
- Extended tech stack: Express, WebSockets, NestJs, MongoDB, Redis, Passport.js, Nodemailer, JWT, Bcrypt, Jest, Pino logger
- Data visualization: ECharts, Konva.js, Elkjs, Dagre

## EXPERIENCE

### Senior Frontend/FullStack Engineer

#### System Solutions Inc

📅 02/2025 -Present 📍 Moscow, Russia

- **Interactive data visualization** and management of business process diagrams and contracts. Creating an in-house library based on Konva (graphics), Elk and Dagre (graph layout)
- **Implementing new design model** with refactoring and integrating best practices (rxjs, standalone, defer import)
- **Performance optimization:** bundle size reduction, lazy loading, data memorization, rendering Detecting and resolving memory leaks
- **Real-Time processing system for stock quotes.** Frequency is up to 50 milliseconds. WebSockets. Page rendering optimization. RxJS streams with buffering and refresh settings. Server states handling
- **Flexible JWT authentication server.** The access + refresh token pattern. Seamless renewal of the expired token. Easily connecting to other applications. The admin panel. Redis caching. Detailed logs

### Senior Full Stack Engineer

#### 3A Solutions Inc

📅 01/2023 - 01/2025 📍 Moscow, Russia

- Developed a decoupled, scalable, ready-to-go **asset management application** platform from scratch.
- Implemented full-scale asset management model.
- Implemented a **comprehensive financial accounting** system.
- **Built RESTful API** architecture using NodeJs (Express) layer.
- **Created frontend** using Angular, Material Design and RxJS. Wrote unit tests using Jest
- Built an effectively structured PostgreSQL DB and created functions to perform complex calculations.



## PROJECTS

- **Business Process & Electronic Document Management System.** A distributed low code platform based on microservice architecture. The system provides tools for flexible business process setup and further tasks management, handling contracts and financial transactions. Global setup allows to conclude and execute inter-domain contracts, between parties, each with its own infrastructure. **Done:**
    - Developed in-house interactive data visualization library. Library covers business processes, contracts diagrams and provides functionality to manage these objects
    - Implemented new UI, design model, best practices, refactored codebase.
    - Contracts and layout modules have been fully rewritten.
    - Bundle size reduced from one chunk 10mb => 5.5 of small chunks => 1.5mb via Brotli compressing. By means of lazy loading application gradually assembles
    - Created widget module
  - **Asset Management Platform.** A full-scale application for financial companies providing investment portfolio management services, including all aspects of investment cycle and accounting  
<https://p2zpsq4w-5001.euw.devtunnels.ms/apps/aam/>  
Node.js REST API
    - More than 100 endpoints to API. Isolated access to the database. Access to third party API (market data)
    - Session authentication via JS passport. Applying DB role to each request by switching query pool
- PostgreSQL:
- 69 tables, 82 functions, 6 group roles, 100 constraints, 13 triggers, 18 RLS policies.
  - Functions: FIFO calculation, Balance open/close/projection, Orders calculation based on model portfolio and applying investment restrictions, Time-weighted rate of return and PL analysis, Investment portfolio detailed statement, Fees calculation (high-water mark, sliding scale fees): management/performance



## EXPERIENCE ...

### Middle Frontend Engineer - 2 years

#### BNP Paribas

📅 03/2021 - 01/2023 📍 Moscow, Russia

- Created BackOffice Dashboard. Flexible UI to get live customized data regarding trades and payments in order to process them duly and timely (Web UI vanilla - Node JS - SQL Server).
- Developed a tax suspension verification service for clients applying for deposits (Browser Automation - Scripting).
- Developed a module to track the document flow for trades' confirmations. User-friendly interface for status management and reconciliation functionality.

### Private Investor

#### Own Asset Portfolio

📅 08/2016 - 02/2021 📍 Moscow, Russia

- Managed my own investment portfolio using various products through brokerage account and collective investments
- Self-education and development in financial markets.

### Head of Operations and Support Department

#### Renaissance Investment Management

📅 02/2006 - 01/2011 📍 Moscow, Russia

- 2 middle office platforms were launched to ensure consecutive growth of the private client asset management business
- Developed an additional module for the first middle-office platform, which allowed it to be launched on time. The module was a crucial part and was constantly being developed in accordance with business demands
- Participated in projects of launching new products/processes: preparation and approval of business requirements, technical specifications, testing plans, maintenance and control of the backlog
- Supported and supervised the day-to-day activities of the department

## EDUCATION

- Practice, documentations, international advanced youtube channels, pet projects
- State University of Management (with honors)
- Central Bank of Russia Banking School (with honors)

## PROJECTS...

- **Stock quotes in real time.** A real-time processing system with a default stream rate of 50 milliseconds. Data exchange via WebSocket. Page rendering optimization. Data stream management using RxJS with buffering and refresh rate settings. Seamless JWT authentication. Extensive handling of server states, errors and notifying user accordingly
- **Flexible JWT authentication server.** The access + refresh token pattern. Seamless renewal of the expired token. Easily connecting to other applications. The admin panel. Redis caching. Detailed logs.
- **Personal CV Site.** Personal site based on a standalone approach. The app includes directives, interceptors, initializers and animation. The site provides my CV data with skills, experience timeline, projects info, about sections. There is also a contact form allowing to get in touch with me by means of telegram api bot
- **BackOffice dashboard.** Dashboard with flexible UI to get live customized data regarding trades and payments in order to process them duly and timely